

Flow A2 - AI Security Content CI for GenAI/MCP/RAG/Agentic Detection-as-Code

GitHub PR-triggered quality gate validating Wazuh rules, MCP policy bundles, RAG/memory controls, agentic policies, schemas, mappings, replay harness, GitHub labels, Slack, and n8n DataTables.

Document Type: Workflow project document | Flow A2

Prototype Focus: AI security content CI | GitHub | n8n | Wazuh | MCP/RAG/Agentic policies

Design Goal: Showcase a production-inspired validation gate for the full GenAI runtime stack before deployment.

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Why this document matters

- The write-up is designed as both a portfolio-grade project document and an implementation-facing prototype note.
- It explains the problem, the workflow logic, the SOC value, and the validation evidence captured from real lab executions.
- The project extends the previous MVP V1 pattern into MCP, RAG/memory, and agentic AI security operations.

Key outcome	Why it matters
PR quality gate	Real GitHub PRs were validated and returned PASS/FAIL comments and labels.
V2 extension	Extends MVP V1 CI beyond prompt/output detection into MCP, RAG/memory, agentic policy, schema, mappings, and regression.
Evidence trail	Slack messages and n8n DataTables preserve CI status, stage results, changed files, and replay rows.

Architecture view

Flow A2 - AI Security Content CI

Validation gate for Wazuh rules, MCP/RAG/agentic policies, mappings, schemas, and replay harness



MVP V2 prototype: production-inspired design, validated in lab, prepared for portfolio and further hardening.

Architecture view - Flow A2 turns a GitHub PR into validated AI security engineering evidence.

Project Summary

Flow A2 is the AI-security content quality gate for MVP V2. It validates pull requests before anything can be deployed by Flow B2. The workflow classifies changed files, runs stage-based validation, comments on GitHub, applies pass/fail labels, posts to Slack, and writes audit rows to n8n DataTables.

Problem Statement

AI security content is now more than one Wazuh XML file. MCP tool policy, RAG/memory policy, agentic approval/identity policy, schema contracts, case-template mapping, and replay expectations can drift together. Without a CI gate, a harmless-looking pull request can break runtime detections or policy enforcement.

Solution Overview

Flow A2 keeps the V1 CI/CD idea but expands the validation scope to the full GenAI runtime stack. Conceptually it is still a detection-as-code gate, but in V2 it validates MCP, RAG/memory, agentic, policy-bundle, schema, TheHive mapping, DataTable, and replay harness integrity.

Tools and Technologies Used

Tool	Role in the prototype
GitHub	Real PR trigger, file listing, comments, labels, and deployment gate metadata.
n8n	Workflow orchestration, branch logic, Code nodes, ExecuteCommand, Slack, and DataTables.
Wazuh content	Rules/decoders are validated before deployment.
Python CI scripts	Stage validators, metadata checks, schema checks, and replay harness execution.
Slack	SOC-facing pass/fail visibility.
n8n DataTables	Persistent audit evidence for CI runs, changed files, stage results, and regression rows.

How the System Works

Stage	What happens
Trigger	A GitHub pull request event opens, syncs, or reopens a PR.
Classify	Changed files are classified into Wazuh, Sigma, MCP, RAG/memory, agentic, schema, mapping, or documentation categories.
Validate	The local CI runner executes 13 stages and returns structured JSON.
Report	The workflow builds GitHub and Slack messages, applies labels, and writes audit rows.
Gate B2	Flow B2 later reads A2 pass labels before deployment is allowed.

SOC Value and Expected Impact

- Expected lower MTTD for broken security content because PR feedback is immediate.
- Improves analyst and detection-engineer consistency by producing the same stage results every time.
- Reduces risky manual deployment by making B2 depend on A2 pass labels.
- Makes AI security policy changes auditable through DataTables and Slack evidence.

Implementation and Validation Evidence

The A2 evidence includes a successful PR validation path and a deliberate fail path. The PASS run validates a valid agentic policy PR; the FAIL run blocks a malformed agentic policy change.

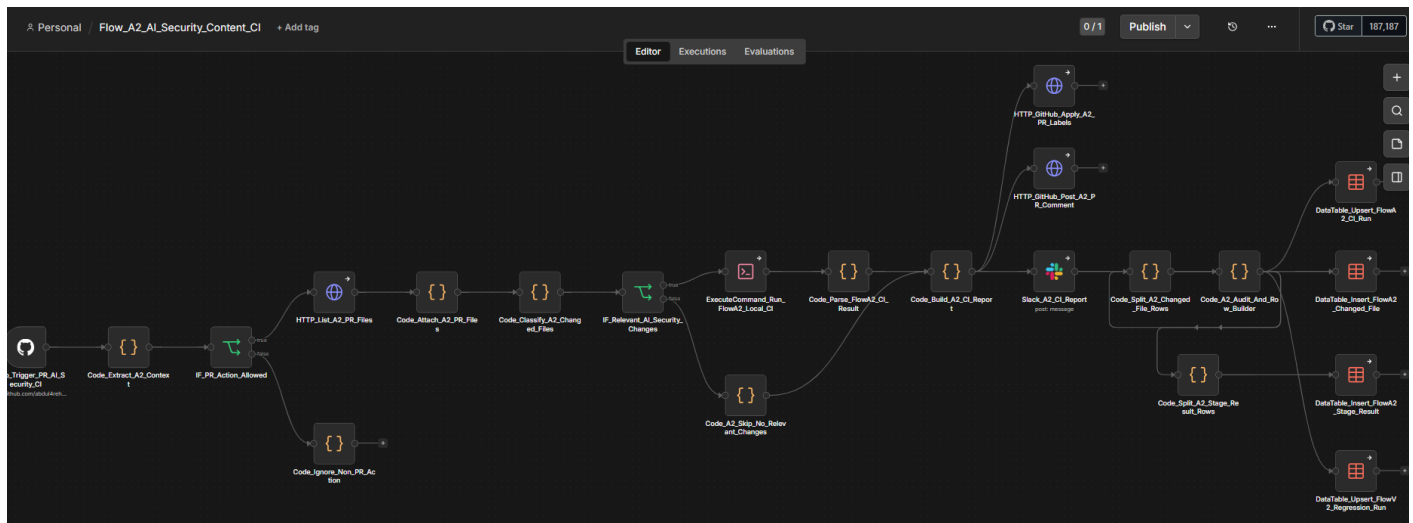


Figure A2-01 - Full Flow A2 production n8n workflow showing GitHub PR trigger, classification, CI execution, reporting, and DataTables.



Figure A2-02 - Left-side A2 workflow view showing GitHub trigger, PR context extraction, file listing, and changed-file classification.

HTTP_List_A2_PR_Files

INPUT

SchemaTableJSONParametersSettings

IF_PR.Action_Allowed1 item

```
{
  "action": "opened",
  "pr_action_allowed": true,
  "owner": "abdu14rehman215",
  "repo_name": "wazuh-genai-ci",
  "repo": "abdu14rehman215/wazuh-genai-ci",
  "pr_number": 19,
  "head_sha": "345e72b7f4abc784d3ea4fb27a653fc016180ff8",
  "branch": "evidence/a2-pass-agentic-policy-fixed",
  "actor": "abdu14rehman215",
  "repo_path": "/opt/detection-ci/wazuh-genai-ci",
  "pr_url": "https://github.com/abdu14rehman215/wazuh-genai-ci/pull/19",
  "received_at": "2026-05-09T10:08:58.773Z",
  "changed_files": [
  ],
  "changed_files_json": "[[]]",
  "changed_files_b64": "Wl0="
}
```

MethodGETURLhttps://api.github.com/repos/{{ \$json.owner }}/{{ \$json.repo_name }}/pulls/{{ \$json.pr_number }}/fileshttps://api.github.com/repos/abdu14rehman215/wazuh-genai-ci/pulls/19/filesAuthenticationPredefined Credential TypeCredential TypeGitHub APISend Query ParametersSend HeadersSpecify HeadersUsing Fields BelowHeadersAcceptNameAccept

OUTPUTExecution will continue even if the node fails1 item

```
{
  "sha": "7f2622c769b0d1709456fbc32968deb6b79d2",
  "filename": "policies/agentic/approval_prompt_policy.json",
  "status": "modified",
  "additions": 1,
  "deletions": 1,
  "changes": 3,
  "blob_url": "https://github.com/abdu14rehman215/wazuh-genai-ci/blob/345e72b7f4abc784d3ea4fb27a653fc016180ff8/policies%2Fagentic%2Fapproval_prompt_policy.json",
  "raw_url": "https://github.com/abdu14rehman215/wazuh-genai-ci/raw/345e72b7f4abc784d3ea4fb27a653fc016180ff8/policies%2Fagentic%2Fapproval_prompt_policy.json",
  "contents_url": "https://api.github.com/repos/abdu14rehman215/wazuh-genai-ci/contents/policies%2Fagentic%2Fapproval_prompt_policy.json?ref=345e72b7f4abc784d3ea4fb27a653fc016180ff8",
  "patch": "@@ -33,5 +33,6 @@\n      \"missing\", \n      \"bypassed\", \n      \"denied\" ] ) + ~ } , \n+     \"evidence_test_note\": \"Flow A2 real GitHub PASS evidence - valid agentic policy metadata note\" \n}"
```

Portfolio prototype documentation | [abdul4rehman215](#)

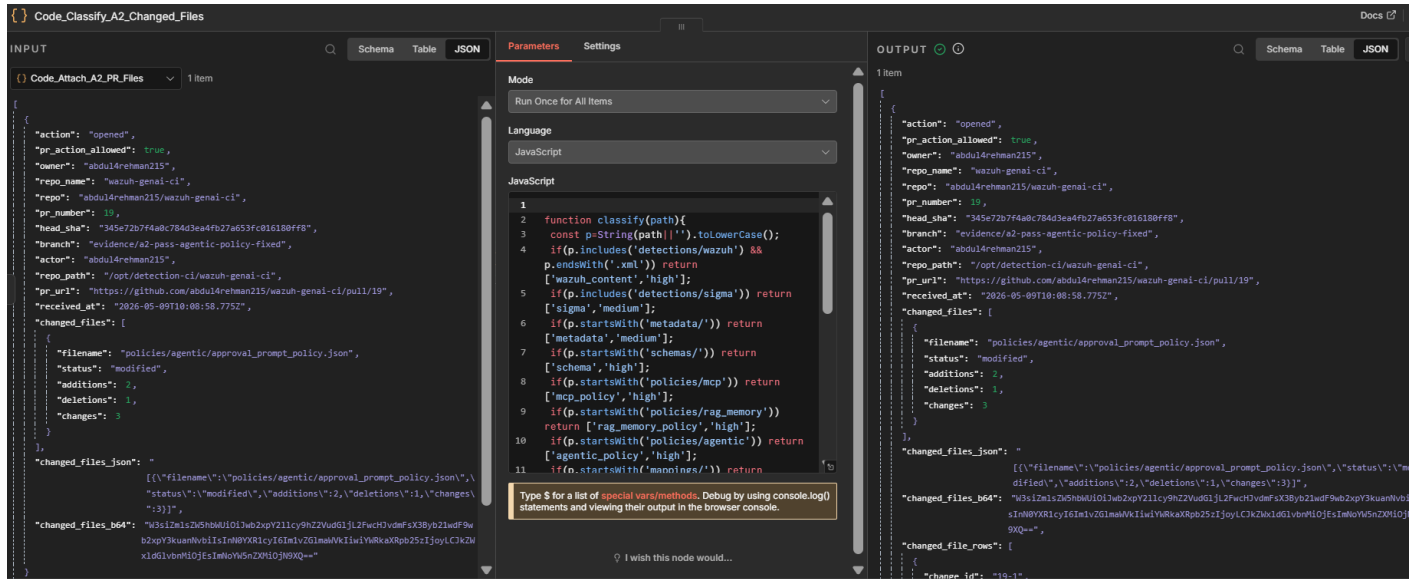


Figure A2-05 - Flow A2 validation node output with structured stage results and workflow metadata.

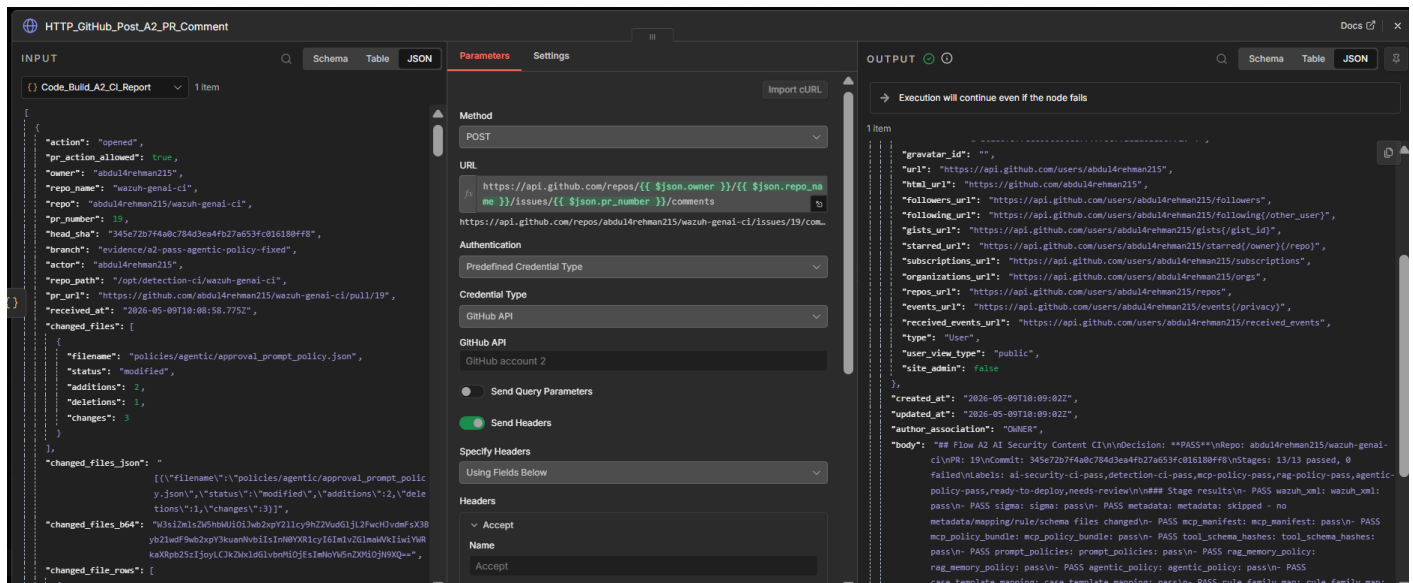


Figure A2-06 - GitHub comment/label API output evidence after the CI result was built.

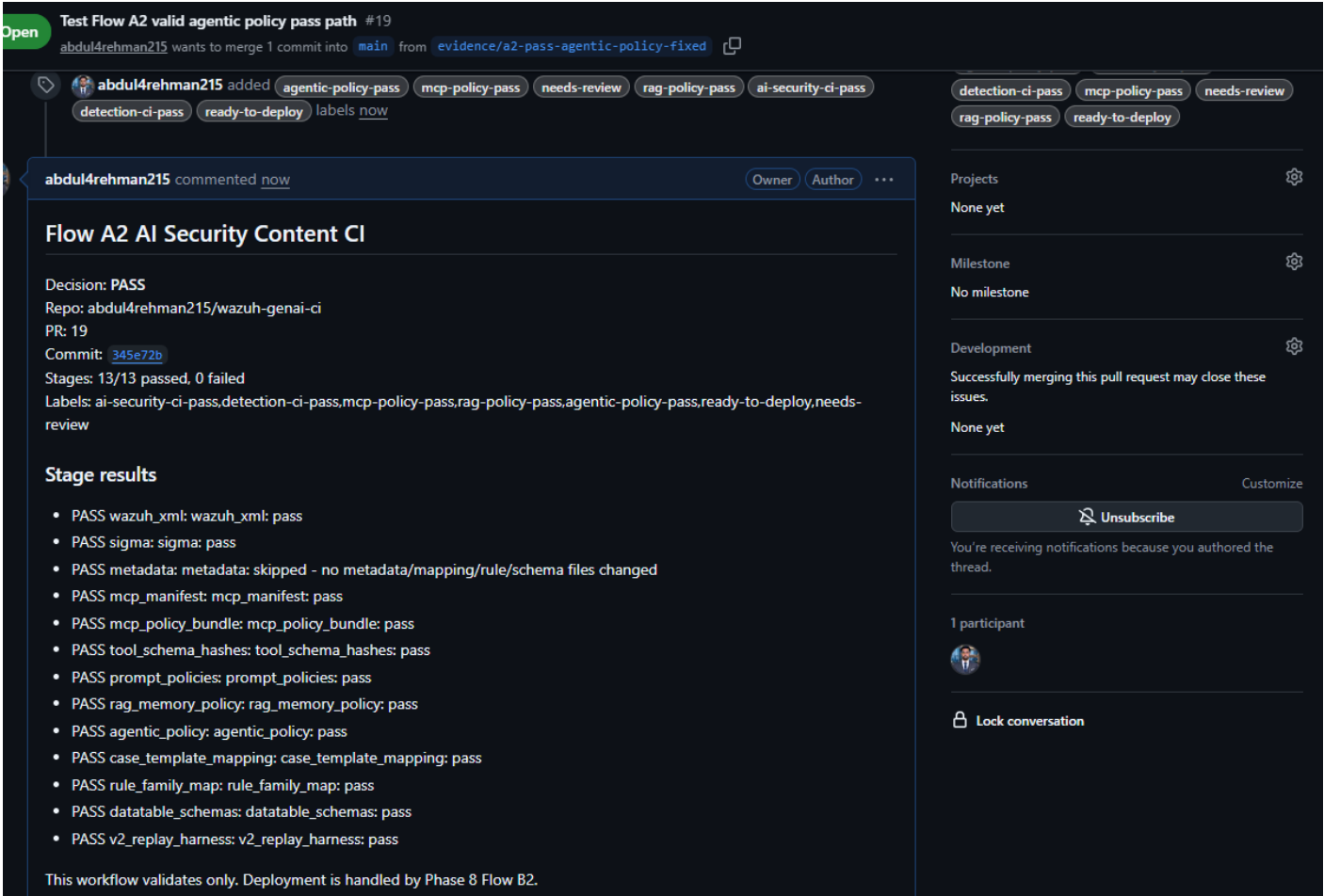


Figure A2-07 - GitHub PR #19 PASS evidence: 13/13 stages passed and V2 pass labels applied.

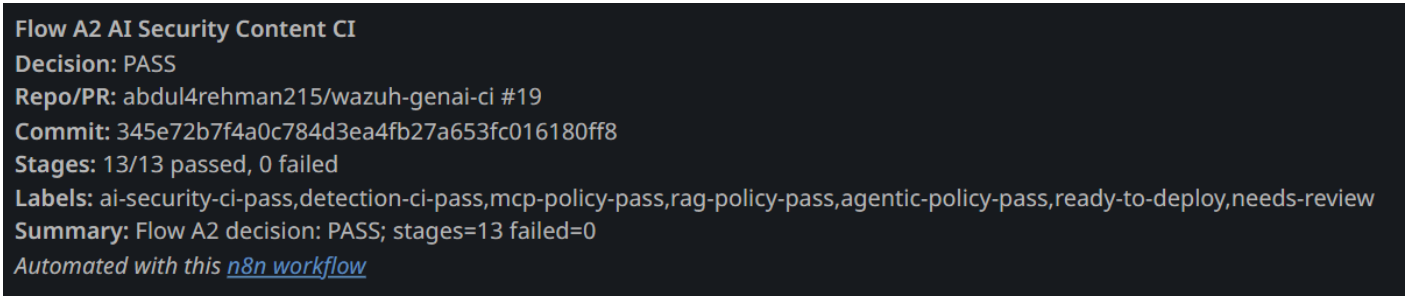


Figure A2-08 - Slack PASS notification for Flow A2 showing decision, repo/PR, commit, and stage counts.

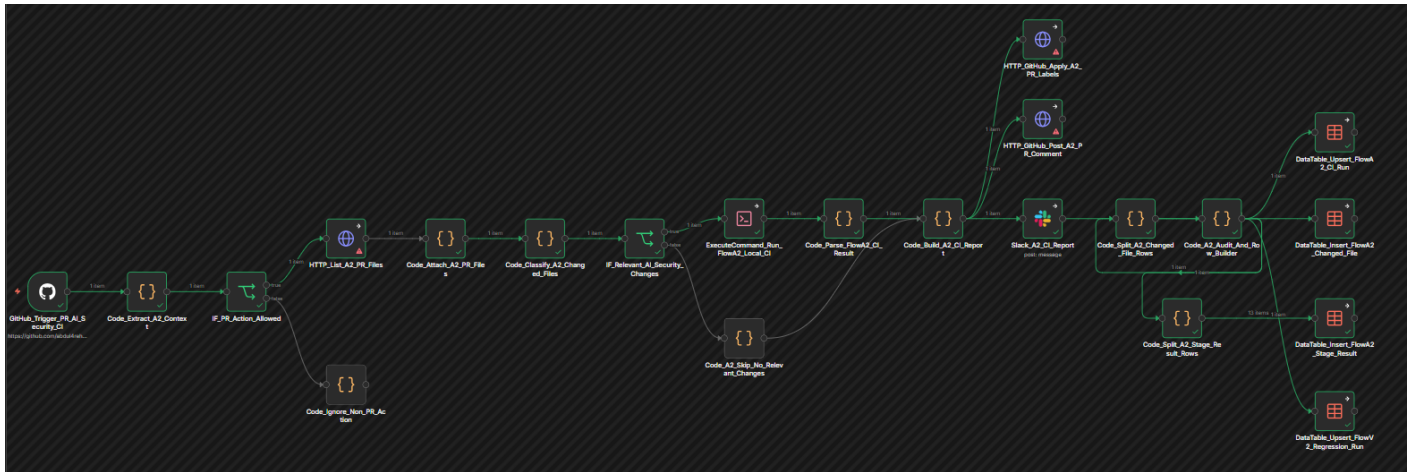


Figure A2-09 - n8n execution canvas for successful A2 run with the complete green execution path.

The screenshot displays the n8n interface for the 'Code_Parse_FlowA2_CI_Result' node. The 'INPUT' section shows the execution context, including the repository name 'wazuh-genai-ci' and the pull request number '20'. The 'PARAMETERS' section shows the 'Mode' set to 'Run Once for All Items' and the 'Language' set to 'JavaScript'. The 'OUTPUT' section shows the parsed JSON data, which includes the 'changed_files' array and the 'changed_files_b64' array. The 'changed_files' array contains details for the file 'policies/agentic/approval_prompt_policy.json', including its status, additions, deletions, and changes. The 'changed_files_b64' array contains the base64-encoded content of the file.

Figure A2-10 - Node-level A2 context and file classification evidence for agentic policy PR validation.

Evidence/a2 fail broken agentic policy #20
 abdul4rehman215 wants to merge 2 commits into `main` from `evidence/a2-fail-broken-agentic-policy`

abdul4rehman215 added `agentic-policy-fail` `ai-security-ci-fail` `detection-ci-pass` `mcp-policy-pass` `needs-review` `rag-policy-pass` labels `now`

abdul4rehman215 commented `now` Owner Author ...

Flow A2 AI Security Content CI

Decision: **FAIL**
 Repo: abdul4rehman215/wazuh-genai-ci
 PR: 20
 Commit: [170c739](#)
 Stages: 12/13 passed, 1 failed
 Labels: ai-security-ci-fail,detection-ci-pass,mcp-policy-pass,rag-policy-pass,agentic-policy-fail,needs-review

Stage results

- PASS wazuh_xml: wazuh_xml: pass
- PASS sigma: sigma: pass
- PASS metadata: metadata: skipped - no metadata/mapping/rule/schema files changed
- PASS mcp_manifest: mcp_manifest: pass
- PASS mcp_policy_bundle: mcp_policy_bundle: pass
- PASS tool_schema_hashes: tool_schema_hashes: pass
- PASS prompt_policies: prompt_policies: pass
- PASS rag_memory_policy: rag_memory_policy: pass
- FAIL agentic_policy: agentic_policy: fail
- PASS case_template_mapping: case_template_mapping: pass
- PASS rule_family_map: rule_family_map: pass
- PASS datatable_schemas: datatable_schemas: pass
- PASS v2_replay_harness: v2_replay_harness: pass

This workflow validates only. Deployment is handled by Phase 8 Flow B2.

Figure A2-11 - GitHub PR #20 FAIL evidence: malformed agentic policy detected and blocked by A2.

Wazuh-SOC-Alerts APP 15:48
Flow A2 AI Security Content CI
Decision: FAIL
Repo/PR: abdul4rehman215/wazuh-genai-ci #20
Commit: 170c739afdb9e4f8ff6f1d9b0cac6d4d00077df8
Stages: 12/13 passed, 1 failed
Labels: ai-security-ci-fail,detection-ci-pass,mcp-policy-pass,rag-policy-pass,agentic-policy-fail,needs-review
Summary: Flow A2 decision: FAIL; stages=13 failed=1
 Automated with this [n8n workflow](#)

Figure A2-12 - Slack FAIL notification for Flow A2 showing agentic-policy failure and needs-review result.

Personal	Data tables	flow_a2_ci_runs	Q Search ⌵ Add Row Add Column								
T branch	# pr_number	T head_sha	T actor	# changed_file_count	# relevant_change_count	# stages_total	# stages_passed	# stages_failed	T decision	T status	T labels_csv
evidence/a2-pass-agentic-policy-fixed	19	345e72b7f4a0c784d3	abdul4rehman215	1	1	13	13	0	pass	passed	ai-security-ci-pass,detection-ci-pass,n
evidence/a2-fail-broken-agentic-policy	20	170c739afdb9e4f8ff6f	abdul4rehman215	1	1	13	12	1	fail	failed	ai-security-ci-fail,detection-ci-pass,mc

Figure A2-13 - flow_a2_ci_runs DataTable with PASS and FAIL audit records.

Data tables
/
flow_a2_ci_changed_files
:

T change_id	T ci_run_id	T file_path	T change_status	T category	T risk_category	T validator_group	☒ requires_review	T notes
a2-change-1	a2-1778321339-19	policies/agentic/approval_prompt_polic	modified	agentic_policy	high	agentic_policy	☒	classified by Flow A2
a2-change-1	a2-1778321899-20	policies/agentic/approval_prompt_polic	modified	agentic_policy	high	agentic_policy	☒	classified by Flow A2

Figure A2-14 - flow_a2_ci_changed_files DataTable showing changed agentic policy classification and review requirement.

A Personal / Data tables / flow_a2_ci_stage_results												
☐	id	T stage_result_id	T ci_run_id	T stage_name	T category	T status	☒	passed	# duration_seconds	# files_checked	# findings_count	T summary
☐	1	a2-1778321339-19-wazuh_xml	a2-1778321339-19	wazuh_xml	wazuh_xml	pass	☒		0.052	6	0	wazuh_xml: pass
☐	2	a2-1778321339-19-sigma	a2-1778321339-19	sigma	sigma	pass	☒		0.044	0	0	sigma: pass
☐	3	a2-1778321339-19-metadata	a2-1778321339-19	metadata	metadata	skip	☒		0	0	0	metadata: skipped - no metadata/mapp
☐	4	a2-1778321339-19-mcp_manifest	a2-1778321339-19	mcp_manifest	mcp_manifest	pass	☒		0.043	1	0	mcp_manifest: pass
☐	5	a2-1778321339-19-mcp_policy_bundle	a2-1778321339-19	mcp_policy_bundle	mcp_policy_bundle	pass	☒		0.041	0	0	mcp_policy_bundle: pass
☐	6	a2-1778321339-19-tool_schema_hashes	a2-1778321339-19	tool_schema_hashes	tool_schema_hashes	pass	☒		0.043	Null		tool_schema_hashes: pass
☐	7	a2-1778321339-19-prompt_policies	a2-1778321339-19	prompt_policies	prompt_policies	pass	☒		0.042	0	0	prompt_policies: pass
☐	8	a2-1778321339-19-rag_memory_policy	a2-1778321339-19	rag_memory_policy	rag_memory_policy	pass	☒		0.041	4	0	rag_memory_policy: pass
☐	9	a2-1778321339-19-agentic_policy	a2-1778321339-19	agentic_policy	agentic_policy	pass	☒		0.105	5	0	agentic_policy: pass
☐	10	a2-1778321339-19-case_template_map	a2-1778321339-19	case_template_mapping	case_template_mapping	pass	☒		0.1	0	0	case_template_mapping: pass
☐	11	a2-1778321339-19-rule_family_map	a2-1778321339-19	rule_family_map	rule_family_map	pass	☒		0.041	0	0	rule_family_map: pass
☐	12	a2-1778321339-19-datatable_schemas	a2-1778321339-19	datatable_schemas	datatable_schemas	pass	☒		0.043	0	0	datatable_schemas: pass
☐	13	a2-1778321339-19-v2_replay_harness	a2-1778321339-19	v2_replay_harness	v2_replay_harness	pass	☒		0.045	0	0	v2_replay_harness: pass
☐	14	a2-1778321899-20-wazuh_xml	a2-1778321899-20	wazuh_xml	wazuh_xml	pass	☒		0.05	6	0	wazuh_xml: pass
☐	15	a2-1778321899-20-sigma	a2-1778321899-20	sigma	sigma	pass	☒		0.041	0	0	sigma: pass
☐	16	a2-1778321899-20-metadata	a2-1778321899-20	metadata	metadata	skip	☒		0	0	0	metadata: skipped - no metadata/mapp
☐	17	a2-1778321899-20-mcp_manifest	a2-1778321899-20	mcp_manifest	mcp_manifest	pass	☒		0.041	1	0	mcp_manifest: pass
☐	18	a2-1778321899-20-mcp_policy_bundle	a2-1778321899-20	mcp_policy_bundle	mcp_policy_bundle	pass	☒		0.041	0	0	mcp_policy_bundle: pass
☐	19	a2-1778321899-20-tool_schema_hashes	a2-1778321899-20	tool_schema_hashes	tool_schema_hashes	pass	☒		0.041	Null		tool_schema_hashes: pass
☐	20	a2-1778321899-20-prompt_policies	a2-1778321899-20	prompt_policies	prompt_policies	pass	☒		0.041	0	0	prompt_policies: pass
☐	21	a2-1778321899-20-rag_memory_policy	a2-1778321899-20	rag_memory_policy	rag_memory_policy	pass	☒		0.04	4	0	rag_memory_policy: pass
☐	22	a2-1778321899-20-agentic_policy	a2-1778321899-20	agentic_policy	agentic_policy	fail	☐		0.044	0	0	agentic_policy: fail
☐	23	a2-1778321899-20-case_template_map	a2-1778321899-20	case_template_mapping	case_template_mapping	pass	☒		0.046	0	0	case_template_mapping: pass
☐	24	a2-1778321899-20-rule_family_map	a2-1778321899-20	rule_family_map	rule_family_map	pass	☒		0.042	0	0	rule_family_map: pass
☐	25	a2-1778321899-20-datatable_schemas	a2-1778321899-20	datatable_schemas	datatable_schemas	pass	☒		0.043	0	0	datatable_schemas: pass
☐	26	a2-1778321899-20-v2_replay_harness	a2-1778321899-20	v2_replay_harness	v2_replay_harness	pass	☒		0.044	0	0	v2_replay_harness: pass

Figure A2-15 - flow_a2_ci_stage_results DataTable showing all validation stages for pass and fail runs.

tables / flow_v2_regression_runs											
started_at	completed_at	trigger_type	repo	head_sha	corpus_version	total_tests	positive_total	positive_passed	positive_failed	negative_total	negative_passed
2026-05-09T15:38:59.	2026-05-09T15:39.	pr	abdul4rehman215/wazu-h-genai-ci	345e72b7f4a0c784d3ea4fb27a653fc0	v2-phase7-local-static	22	19	19	0	3	3
2026-05-09T15:48:19.	2026-05-09T15:48.	pr	abdul4rehman215/wazu-h-genai-ci	170c739afdb9e4f8ff6fd9b0cac6d4d0f	v2-phase7-local-static	22	19	19	0	3	3

Figure A2-16 - flow_v2_regression_runs / replay evidence row linked back to Flow A2 CI validation.

Limitations and Production Hardening Roadmap

- The workflow is an MVP lab prototype and still needs production-grade GitHub App authentication, branch protection integration, and signed commit enforcement.
- The current stage set is practical but should be expanded with additional negative corpora, schema fuzzing, policy mutation tests, and secret scanning.
- Production use should add retention controls, immutable audit storage, and metrics tied to real incident outcomes.

Skills Demonstrated

Skill area	Evidence in this build
Detection engineering CI	Implemented PR classification, staged validation, and replay-aware pass/fail logic.
Workflow automation	Built a multi-branch n8n workflow connecting GitHub, shell validation, Slack, and DataTables.
AI security governance	Validated MCP, RAG/memory, and agentic policy surfaces.
Operational evidence	Captured GitHub, Slack, n8n, and DataTable proof for portfolio and SOC auditability.

Conclusion

Flow A2 proves that MVP V2 is not only a runtime demo; it has an engineering control plane. It validates AI security content before deployment and creates evidence that Flow B2 can trust.

This is the first production-style layer of the V2 prototype: content quality, repeatability, and deploy readiness.

Document prepared for portfolio showcase by [abdul4rehman215](#)